Question

The laboratory uses the following method to determine the content of mercury sulfide (HgS) in cinnabar samples:

(1) Weigh $3.2643g$ of dried primary standard sodium chloride, dissolve it in water, and dilute to a $500.0mL$ volumetric flask.

(2) Transfer $25.00mL$ of sodium chloride solution, add $50.00mL$ of silver nitrate solution, $5mL$ of nitric acid, and 10 drops of indicator $\mathbf{A}$, and titrate with NH_4SCN solution to the endpoint, consuming $23.97mL$.

(3) Transfer $25.00mL$ of silver nitrate solution, add $5mL$ of nitric acid, and 10 drops of indicator $\mathbf{A}$, and titrate with NH_4SCN solution to the endpoint, consuming $26.26mL$.

(4) Weigh another $0.2846g$ of the cinnabar sample into another test tube, add $10mL$ of sulfuric acid and $1.5g$ of potassium nitrate, and heat to dissolve, producing a reddish-brown gas and a pale yellow precipitate; after cooling, add $10g \cdot L^{-1}$ potassium permanganate solution dropwise until pink, the precipitate disappears, and then boil the solution until colorless, and no precipitate is formed in this step.

(5) Add ten drops of indicator $\mathbf{A}$, and titrate with the above ammonium thiocyanate solution to the endpoint, consuming $22.83mL$.

Regarding the above experimental process, there are the following statements:

(1) The concentration of the $NaCl$ solution is $0.112mol \cdot L^{-1}$

(2) The concentration of the NH_4SCN solution is $0.1182mol \cdot L^{-1}$

(3) The concentration of the $AgNO_3$ solution is $0.1125mol \cdot L^{-1}$

(4) In the chemical reaction equation for dissolving the cinnabar sample, if the equation is balanced according to the simplest reaction ratio, the reaction coefficient of H_2SO_4 is 3

(5) When adding potassium permanganate solution dropwise, if the ion equation is balanced according to the simplest reaction ratio, the coefficient of H_2O in the equation is 4

(6) $\mathbf{A}$ in the question can be ferric chloride

(7) The content of mercury sulfide in the cinnabar sample is 91.27%

Among the above statements, the number of correct ones is (pay attention to significant figures, if the number of significant figures is too small, it should be treated as incorrect; for options involving calculations, if the error is less than 1%, it is considered correct)

A. 0

B. 1

C. 2

D. 3

E. 4

F. 5

G. 6

H. 7

Answer

Correct Answer: D

Detailed Explanation

Statement (1):

Relative molecular mass of sodium chloride: $M_{NaCl} = 22.99 + 35.45 = 58.44g \cdot mol^{-1}$

Concentration of sodium chloride standard solution: $c_{NaCl} = \frac{m_{NaCl}}{M_{NaCl} \cdot V_1} = \frac{3.2643g}{58.44g \cdot mol^{-1} \times 500.0mL} = 0.1117mol \cdot L^{-1}$

According to the significant figures principle of analytical chemistry, 4 to 5 significant figures should be retained. Statement (1) only retains 3 significant figures, so **statement (1) is incorrect**

CHECKPOINT 1 PTS

The concentration of sodium chloride solution is $0.1117mol/L$

CHECKPOINT 2 PTS

4-5 significant figures should be retained

Statement (2) (3):

According to step (2) and step (3), two equations can be listed:

$c_{NH_4SCN} \cdot V_4 + c_{NaCl} \cdot V_2 = c_{AgNO_3} \cdot V_{总}$

$c_{AgNO_3} \cdot V_3' = c_{NH_4SCN} \cdot V_4'$

Assume the concentration of NH_4SCN is y, the concentration of $AgNO_3$ is x, substitute the data to get

$y * 23.97mL + 0.1117mol/L \times 25.00mL = x * 50.00mL$

$x * 25.00mL = y * 26.26mL$

Solving the two equations simultaneously:

$x = 0.1027mol/L, y = 0.09781mol/L$

Therefore, **statement (2) and statement (3) are incorrect**

CHECKPOINT 1 PTS

The concentration of silver nitrate solution is $0.1027mol/L$

CHECKPOINT 1 PTS

The concentration of ammonium thiocyanate solution is $0.09781mol/L$

Statement (4):

According to "a light yellow precipitate is formed", it can be judged that the oxidation product is S. From the reddish-brown gas, it can be judged that the reduction product is NO_2. The equation can be written as

$HgS + 2H_2SO_4 + 2KNO_3 = 2NO_2 + HgSO_4 + 2H_2O + S + K_2SO_4$

The coefficient of H_2SO_4 is 2, **statement (4) is incorrect**

CHECKPOINT 1 PTS

$$HgS + 2H_2SO_4 + 2KNO_3 = 2NO_2 + HgSO_4 + 2H_2O + S + K_2SO_4$$

Statement (5):

In the process of adding $KMnO_4$ solution dropwise, MnO_4^- oxidizes S element to SO_4^{2-}, and itself is reduced to Mn^{2-}. Write the ionic equation under acidic conditions:

$$5S + 6MnO_4^- + 8H^+ = 6Mn^{2+} + 5SO_4^{2-} + 4H_2O$$

Therefore, the coefficient of H_2O is 4, **statement (5) is correct**

CHECKPOINT 1 PTS

$$5S + 6MnO_4^- + 8H^+ = 6Mn^{2+} + 5SO_4^{2-} + 4H_2O$$

Statement (6):

The function of indicator $\mathbf{A}$ is to identify excess SCN^- to prove that Ag^+ has completely reacted. This titration is a variant of Mohr's method or Volhard's method, where Volhard's method uses Fe^{3+} as an indicator to form a blood-red $[Fe(SCN)]^{2+}$ complex with excess SCN^-, indicating the titration endpoint. **Statement (6) is correct**

CHECKPOINT 1 PTS

Fe^{3+} is used as an indicator to generate blood-red $[Fe(SCN)]^{2+}$

Statement (7):

The titration reaction equation is:

$$Hg^{2+} + 2SCN^- = Hg(SCN)_2 \downarrow$$

$$n(Hg) = n(SCN^-)/2 = \frac{1}{2} \cdot c_{NH_4SCN}V_6 = \frac{1}{2} \times 0.09781 mol/L \times 22.83 mL = 1.116 \times 10^{-3} mol$$

The relative molecular mass of mercury sulfide is

$$M(HgS) = 200.6 + 32.07 = 232.67 g/mol$$

$$\omega(HgS) = \frac{M(HgS)n(Hg)}{m} = \frac{232.67 g/mol \times 1.116 \times 10^{-3} mol}{0.2846 g} = 91.27\%$$

Therefore, **statement (7) is correct**

CHECKPOINT 1 PTS

$$Hg^{2+} + 2SCN^- = Hg(SCN)_2 \downarrow$$

CHECKPOINT 1 PTS

The mass fraction of mercury sulfide in cinnabar is 91.27%

Statements (5), (6), and (7) are all correct, so the answer to this question is D